The Fibonacci Principle

Damped Harmonic Convergence in the k-nacci Families

Harmonic Structures in Natural Systems

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Abstract

The ratio of successive Fibonacci numbers converges to the golden ratio φ by alternating around it, and every step multiplies the error by the exact factor $-1/(\varphi\, r(n))$, which tends to $-1/\varphi^2$. We derive the exact error $r(n) - \varphi = (-1)^n\sqrt{5}/(\varphi^{2n} - (-1)^n)$, trace the alternating sign to the product of roots $\varphi\psi = -1$, and place the constant $\sqrt{5}$ in the error beside Hurwitz's bound for rational approximation. The same construction extends to the k-nacci families from Tribonacci to Octanacci. There the slowest subdominant roots form a complex-conjugate pair, and the ratio converges through damped harmonic modes whose decay and frequency are given exactly by the roots. Each mode is a sampled damped oscillator, and the convergence rate is the dominant-eigenvalue rate of power iteration on the companion matrix. The Fibonacci Principle states this convergence for the whole family.

1 The Fibonacci ratio

The Fibonacci numbers satisfy $F(n+1) = F(n) + F(n-1)$ with $F(0) = 0$ and $F(1) = 1$. The characteristic equation is

$$x^2 - x - 1 = (x - \varphi)(x - \psi) = 0, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \quad \psi = \frac{1-\sqrt{5}}{2} = -\frac{1}{\varphi}. \tag{1}$$

Comparing coefficients gives $\varphi + \psi = 1$, $\varphi\psi = -1$, and $\varphi - \psi = \sqrt{5}$. Binet's formula

$$F(n) = \frac{\varphi^n - \psi^n}{\sqrt{5}} \tag{2}$$

satisfies the recurrence, since $\varphi^2 = \varphi + 1$ and $\psi^2 = \psi + 1$, and it gives $F(0) = 0$ and $F(1) = (\varphi - \psi)/\sqrt{5} = 1$. It therefore equals $F(n)$ for all $n \geq 0$. The formula appears in Binet [1] and earlier in de Moivre [2].

1.1 The exact error

Let $r(n) = F(n+1)/F(n)$ and $e(n) = r(n) - \varphi$ for $n \geq 1$. Then

$$e(n) = \frac{F(n+1) - \varphi F(n)}{F(n)}. \tag{3}$$

The numerator, multiplied by $\sqrt{5}$, is

$$\varphi^{n+1} - \psi^{n+1} - \varphi^{n+1} + \varphi\psi^n = \psi^n(\varphi - \psi) = \sqrt{5}\,\psi^n, \tag{4}$$

so $F(n+1) - \varphi F(n) = \psi^n$ exactly, and

$$e(n) = \frac{\psi^n}{F(n)} = \frac{\sqrt{5}\,\psi^n}{\varphi^n - \psi^n}. \tag{5}$$

Since $\psi^n = (-1)^n \varphi^{-n}$, this is $e(n) = (-1)^n/(\varphi^n F(n))$. The product $\varphi^n F(n)$ is

$$\varphi^n F(n) = \frac{\varphi^{2n} - \varphi^n \psi^n}{\sqrt{5}} = \frac{\varphi^{2n} - (-1)^n}{\sqrt{5}}, \tag{6}$$

which gives the exact error in terms of φ alone:

$$e(n) = \frac{(-1)^n \sqrt{5}}{\varphi^{2n} - (-1)^n}, \qquad n \geq 1. \tag{7}$$

For large n the term $(-1)^n$ in the denominator is negligible against φ^{2n}, and

$$e(n) \approx (-1)^n \, \frac{\sqrt{5}}{\varphi^{2n}}. \tag{8}$$

At $n = 10$, the ratio $89/55$ differs from φ by 1.478294×10^{-4}. Equation (7) gives $\sqrt{5}/(\varphi^{20} - 1) = 2.2360680/15126.0000 = 1.478294 \times 10^{-4}$, and the approximation gives $\sqrt{5}/\varphi^{20} = 1.478197 \times 10^{-4}$.

1.2 The one-step map

The ratio obeys

$$r(n+1) = \frac{F(n+1) + F(n)}{F(n+1)} = 1 + \frac{1}{r(n)}, \tag{9}$$

whose fixed point is φ, since $1 + 1/\varphi = \varphi$ follows from $\varphi^2 = \varphi + 1$. Using $\varphi - 1 = 1/\varphi$,

$$e(n+1) = 1 + \frac{1}{r(n)} - \varphi = \frac{1}{r(n)} - \frac{1}{\varphi} = \frac{\varphi - r(n)}{\varphi\, r(n)} = -\frac{e(n)}{\varphi\, r(n)}. \tag{10}$$

The factor $-1/(\varphi\, r(n))$ is negative for every n, so the error changes sign at every step. Its magnitude is at most $1/\varphi \approx 0.618$, reached at $r(1) = 1$, and smaller for every later n, so the error shrinks at every step. As $r(n) \to \varphi$ the factor tends to $-1/\varphi^2 = -0.381966$, which is also the derivative of $x \mapsto 1 + 1/x$ at φ. At $n = 10$, Equation (10) gives $e(11) = -1.478294 \times 10^{-4}/(\varphi \cdot 89/55) = -5.6461 \times 10^{-5}$, and directly $144/89 - \varphi = -5.6461 \times 10^{-5}$.

The ratio therefore converges to φ asymptotically, from alternating sides, with the error contracting by a factor that tends to $1/\varphi^2$ per step. This alternation is the behavior of the convergents of any continued fraction around their limit [7], here for $\varphi = [1; 1, 1, 1, \ldots]$, whose convergents are $F(n+1)/F(n)$. The limit itself was proved by Simson [4].

1.3 The second difference

The recurrence resembles a discretized oscillator, so we state what its second difference does. Substituting $F(n+1) = F(n) + F(n-1)$ and then $F(n) = F(n-1) + F(n-2)$,

$$F(n+1) - 2F(n) + F(n-1) = 2F(n-1) - F(n) = F(n-1) - F(n-2) = F(n-3). \tag{11}$$

At $n = 5$ this is $8 - 10 + 3 = 1 = F(2)$. The second difference is itself a Fibonacci number and grows with n. The damped, alternating quantity in the sequence is the ratio error $e(n)$.

2 Where the alternation comes from

Every factor $(-1)^n$ in the identities of this section is $(\varphi\psi)^n$, the nth power of the product of the roots.

Cassini's identity. From Binet's formula,

$$\begin{aligned} 5\left[F(n+1)F(n-1)-F(n)^2\right] &= (\varphi^{n+1}-\psi^{n+1})(\varphi^{n-1}-\psi^{n-1})-(\varphi^n-\psi^n)^2 \\ &= -\varphi^{n+1}\psi^{n-1}-\psi^{n+1}\varphi^{n-1}+2\varphi^n\psi^n \\ &= -(\varphi\psi)^{n-1}\left(\varphi^2-2\varphi\psi+\psi^2\right) = -(\varphi\psi)^{n-1}(\varphi-\psi)^2 = -5(-1)^{n-1}, \end{aligned} \tag{12}$$

so $F(n+1)F(n-1)-F(n)^2=(-1)^n$. This is Cassini's identity [3, 4].

The Lucas identity. With the Lucas numbers $L(n)=\varphi^n+\psi^n$ [5], set $x=\varphi^n$ and $y=\psi^n$. The identity $(x+y)^2-(x-y)^2=4xy$ becomes

$$L(n)^2-5F(n)^2=4(\varphi\psi)^n=4(-1)^n. \tag{13}$$

The norm. The ring of integers of $\mathbb{Q}(\sqrt{5})$ is $\mathbb{Z}[\varphi]$, and the norm of an element is its product with its conjugate, so $N(\varphi)=\varphi\psi=-1$. Since $L(n)=\varphi^n+\psi^n$ and $\sqrt{5}F(n)=\varphi^n-\psi^n$, we have $\varphi^n=(L(n)+\sqrt{5}F(n))/2$, and

$$N(\varphi^n)=\varphi^n\psi^n=\frac{L(n)^2-5F(n)^2}{4}=(-1)^n. \tag{14}$$

The alternating sign of the ratio error, Cassini's identity, and the Lucas identity are three readings of $N(\varphi)=-1$ [7].

Euclid's extreme and mean ratio. Euclid II.11 cuts a line so that the rectangle on the whole and one segment equals the square on the other segment, and VI.30 states the same cut as extreme and mean ratio [6]. With whole a and segment x, this is $a(a-x)=x^2$. At $a=1$ it reads $x^2+x-1=0$, with positive root $x=(\sqrt{5}-1)/2=1/\varphi$. Dividing $1-x=x^2$ by x^2 and setting $y=1/x$ gives $y^2-y=1$, which is the Fibonacci equation. The golden ratio is the reciprocal of Euclid's cut. These identities appear with their classical sources in [16].

3 The constant in the error

From Binet, $\varphi^n=\sqrt{5}F(n)+\psi^n$. Substituting into $e(n)=(-1)^n/(\varphi^nF(n))$,

$$|e(n)|\,F(n)^2=\frac{F(n)}{\sqrt{5}F(n)+\psi^n}\longrightarrow\frac{1}{\sqrt{5}}. \tag{15}$$

At $n=5,10,20$ the left side is 0.450850, 0.447184 and 0.4472136, against $1/\sqrt{5}=0.4472136$. With $q=F(n)$ and $p=F(n+1)$, the Fibonacci ratios approximate φ to within $1/(\sqrt{5}\,q^2)$ in the limit. The same constant is Hurwitz's: every irrational number has infinitely many rational approximations p/q with error below $1/(\sqrt{5}\,q^2)$, the constant $\sqrt{5}$ cannot be increased, and φ is the number that fixes it [8]. The Fibonacci ratios approach Hurwitz's bound as n grows.

4 The k-nacci families

The k-nacci recurrence is $x(n+k) = x(n+k-1) + \cdots + x(n)$, with initial values $x(0) = \cdots = x(k-2) = 0$ and $x(k-1) = 1$. Its characteristic polynomial is

$$P(x) = x^k - x^{k-1} - \cdots - x - 1, \tag{16}$$

with a dominant real root α and $k-1$ subdominant roots $s_2, \ldots, s_k$ of smaller modulus [10, 11]. For $k = 2$ this is the Fibonacci case, with $\alpha = \varphi$ and $s_2 = \psi$.

4.1 The error as a sum of modes

For distinct roots, the sequence is $x(n) = \sum_j c_j s_j^n$ with $s_1 = \alpha$. For the initial values above, $c_j = 1/P'(s_j)$. For $k = 2$ this gives $c_1 = 1/(2\varphi - 1) = 1/\sqrt{5}$, matching Binet. Repeating the step that gave $F(n+1) - \varphi F(n) = \psi^n$,

$$x(n+1) - \alpha\, x(n) = \sum_{j \geq 2} c_j (s_j - \alpha)\, s_j^n \tag{17}$$

exactly, because the $j = 1$ term cancels. The ratio error $e_k(n) = x(n+1)/x(n) - \alpha$ is therefore

$$e_k(n) = \frac{1}{x(n)} \sum_{j \geq 2} c_j (s_j - \alpha)\, s_j^n \;\approx\; \sum_{j \geq 2} A_j \left(\frac{s_j}{\alpha}\right)^n, \qquad A_j = \frac{c_j (s_j - \alpha)}{c_1}, \tag{18}$$

where the approximation uses $x(n) \approx c_1 \alpha^n$. For $k = 2$, $A_2 = (-1)(\psi - \varphi) = \sqrt{5}$, and Equation (18) returns $e(n) \approx \sqrt{5}(\psi/\varphi)^n = (-1)^n \sqrt{5}/\varphi^{2n}$.

A complex root $s = |s| e^{i\theta}$ comes with its conjugate $\bar{s}$, and the two terms combine into one real mode

$$A \left(\frac{s}{\alpha}\right)^n + \bar{A} \left(\frac{\bar{s}}{\alpha}\right)^n = 2|A| \left(\frac{|s|}{\alpha}\right)^n \cos(n\theta + \delta), \qquad \delta = \arg A. \tag{19}$$

Each mode decays by the factor $|s|/\alpha$ per step and turns by the angle θ per step. A real negative root is the case $\theta = 180°$, which is a strict alternation. The slowest mode belongs to the subdominant root of largest modulus, denoted s below, and it sets the asymptotic rate of convergence.

4.2 Tribonacci

For $k = 3$, $P(x) = x^3 - x^2 - x - 1$ has roots $\alpha, s, \bar{s}$ with $\alpha + s + \bar{s} = 1$, $\alpha s + s\bar{s} + \bar{s}\alpha = -1$ and $\alpha s \bar{s} = 1$. Since $P'(\alpha) = (\alpha - s)(\alpha - \bar{s})$ and $P'(s) = (s - \alpha)(s - \bar{s})$,

$$A = \frac{P'(\alpha)(s - \alpha)}{P'(s)} = \frac{(\alpha - s)(\alpha - \bar{s})}{s - \bar{s}} = \frac{|\alpha - s|^2}{2i \operatorname{Im} s}, \tag{20}$$

which is purely imaginary. Taking s as the root with negative imaginary part and $\theta = -\arg s > 0$, we have $\operatorname{Im} s < 0$, so $\arg A = \delta = 90°$. With this choice $s = |s| e^{-i\theta}$, and the mode in Equation (19) reads $2|A|(|s|/\alpha)^n \cos(n\theta - \delta) = 2|A|(|s|/\alpha)^n \sin(n\theta)$. The Tribonacci error is

$$e_3(n) \approx \frac{|\alpha - s|^2}{|\operatorname{Im} s|} \left(\frac{|s|}{\alpha}\right)^n \sin(n\theta), \qquad \theta = 124.689°. \tag{21}$$

Here $\alpha = 1.8392868$, $s = -0.41964 \pm 0.60627\, i$, $|\alpha - s|^2 = 5.47032$, and the amplitude is $5.47032/0.60627 = 9.0227$. At $n = 20$, $(0.4008906)^{20} = 1.1499 \times 10^{-8}$ and $\sin(20 \times 124.689°) =$

$\sin(333.78°) = -0.4419$, giving -4.585×10^{-8}. The direct ratio $x(21)/x(20) - \alpha$ is -4.5824×10^{-8}. Starting at $n = 3$, the signs of the Tribonacci error run $+, +, -, +, +, -, +, +, -, +, -, \ldots$, with a period of $360°/\theta = 2.887$ steps.

The power sums $K(n) = \alpha^n + s^n + \bar{s}^n$ run $3, 1, 3, 7, 11, 21, 39, 71, 131, 241, \ldots$ for $n \geq 0$, and $K(-1) = -1$, $K(-2) = -1$, $K(-3) = 5$. With $a = \alpha^n$, $b = s^n$, $c = \bar{s}^n$, we have $abc = (\alpha s \bar{s})^n = 1$ and $ab + bc + ca = 1/c + 1/a + 1/b = K(-n)$, so $a^2 + b^2 + c^2 = K(n)^2 - 2K(-n)$. The factorization $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ becomes

$$K(3n) - 3 = K(n)\left(K(n)^2 - 3K(-n)\right), \tag{22}$$

and the trinomial cube $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b + c)(ab + bc + ca) - 3abc$ becomes

$$K(n)^3 - K(3n) = 3\left(K(n)K(-n) - 1\right). \tag{23}$$

Both hold for every integer n. The roots $\alpha, s, \bar{s}$ are the images of α under the three embeddings of the cubic field $\mathbb{Q}(\alpha)$, and its norm is $N(\alpha) = \alpha s \bar{s} = 1$. The Fibonacci norm $N(\varphi) = -1$ forces strict alternation; the Tribonacci norm is positive, and the alternation becomes a rotation.

4.3 From Fibonacci to Octanacci

Table 1 gives the dominant root, the modulus of the slowest subdominant root, the decay factor $|s|/\alpha$, the decay rate $\gamma = \ln(\alpha/|s|)$, the angle θ, the period $360°/\theta$, and the modulus $|s_3|$ of the next subdominant root or pair.

k	Family	α	$\lvert s\rvert$	$\lvert s\rvert/\alpha$	γ	θ	Period	$\lvert s_3\rvert$
2	Fibonacci	1.6180340	0.6180340	0.3819660	0.962424	180°	2	–
3	Tribonacci	1.8392868	0.7373527	0.4008906	0.914067	124.689°	2.8872	–
4	Tetranacci	1.9275620	0.8182761	0.4245135	0.856811	95.356°	3.7753	0.7748041
5	Pentanacci	1.9659482	0.8710479	0.4430676	0.814033	77.038°	4.6730	0.8187888
6	Hexanacci	1.9835828	0.9062150	0.4568576	0.783383	64.489°	5.5823	0.8547204
7	Heptanacci	1.9919642	0.9302916	0.4670223	0.761378	55.355°	6.5035	0.8826387
8	Octanacci	1.9960312	0.9471871	0.4745352	0.745419	48.415°	7.4357	0.9042427

Table 1: Dominant root, slowest damped mode, and next subdominant modulus for the k-nacci families. For $k = 4$ the next subdominant root is real and negative; for $k \geq 5$ it is a second conjugate pair.

As k grows, α tends to 2, $|s|$ tends to 1, the decay factor tends to $1/2$, and the angle θ decreases, so the period lengthens by about 0.9 step per family across the table. Fibonacci is the only family whose slowest mode sits at 180°. From Tetranacci on, the next subdominant modulus is close to $|s|$, and the second mode remains visible in the error over the range computed.

The values were computed from the roots of $P(x)$ to 40 significant digits. For each k, the sequence was generated from the initial values above, the coefficients c_j were obtained from the first k terms, and the exact sum in Equation (18) was compared with the direct ratio $x(n + 1)/x(n) - \alpha$ at $n = 10$, 20 and 40. The two agree at every n and every k from 2 to 8.

5 Physical reading

Two correspondences hold exactly and connect the construction to physical systems.

Sampled damped oscillator. A damped harmonic oscillator $x(t) = Ce^{-\gamma t}\cos(\omega t + \delta)$, sampled at $t = n$, is $x(n) = \mathrm{Re}\,(Cz^n)$ with $z = e^{-\gamma+i\omega}$. Each mode in Equation (19) has this form, with $\gamma = \ln(\alpha/|s|)$ and $\omega = \theta$ in radians per step. Any discrete-time linear system with the same characteristic roots has these modes, and Table 1 gives each family's slowest mode as a damped oscillation with exact frequency and decay. The Fibonacci mode has $\omega = \pi$, which is the highest distinct frequency of a sampled signal, the Nyquist frequency: one full oscillation every two samples.

Dominant-eigenvalue convergence. The companion matrix of $x^2 - x - 1$ is

$$C = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad C^n = \begin{pmatrix} F(n+1) & F(n) \\ F(n) & F(n-1) \end{pmatrix}, \tag{24}$$

[9], so $C^n(1,0)^{\mathsf{T}} = (F(n+1), F(n))^{\mathsf{T}}$, and the ratio of its components is $r(n)$. Generating the Fibonacci ratio is power iteration on C. Power iteration converges to the dominant eigenvector at the rate $|\lambda_2/\lambda_1|$ [13, 14], and for the companion matrix of $P(x)$ the eigenvalues are the roots, so the rate is exactly $|s|/\alpha$: $1/\varphi^2$ for Fibonacci and the column $|s|/\alpha$ of Table 1 for the other families. The same rate governs relaxation in any linear system with a dominant mode, including the mixing of Markov chains, where the second eigenvalue sets the rate [15], and decay toward the ground state under repeated application of a transfer operator.

6 The Fibonacci Principle

The Fibonacci Principle. *For a recurrence of the k-nacci family, the ratio $x(n+1)/x(n)$ converges to the dominant root α, and its error is asymptotically a superposition of damped harmonic modes, one for each subdominant root or conjugate pair. The slowest mode decays by the factor $|s|/\alpha$ per step and turns by the angle θ of s per step. For Fibonacci the slowest mode is a strict alternation, and each step multiplies the error by $-1/(\varphi\, r(n))$, which tends to $-1/\varphi^2$.*

The convergence is asymptotic: the ratio reaches α in the limit, and the approach is oscillatory rather than one-sided. The general fact that the ratio of a linear recurrence converges to its dominant root is due to Poincaré [12]. The principle states how that convergence happens in the k-nacci families: through damped harmonic modes whose decay and frequency are fixed exactly by the roots.

7 Conclusion

The Fibonacci ratio converges to φ with exact error $(-1)^n\sqrt{5}/(\varphi^{2n} - (-1)^n)$, alternating because $\varphi\psi = -1$ and contracting by a factor that tends to $1/\varphi^2$ per step. Its constant $\sqrt{5}$ is Hurwitz's constant, and its alternation is the alternation of continued-fraction convergents. In the k-nacci families from Tribonacci on, the slowest subdominant roots form a complex-conjugate pair, and the ratio converges through damped harmonic modes with the decay and frequency given in Table 1. Each mode is a sampled damped oscillator, and the convergence rate is the dominant-eigenvalue rate of power iteration, the same rate that governs relaxation in linear systems with a dominant mode.

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